Curriculum Vitæ
Hans Halvorson

Stuart Professor of Philosophy, Princeton University

Department of Philosophy, Princeton University
Princeton, NJ 08544
hanshalvorson.com

Appointments

2016– Stuart Professor of Philosophy, Princeton University
2012– associated faculty, Department of Mathematics, Princeton University
2010– professor of philosophy, Princeton University
2005–10 associate professor, Princeton University
2001–05 assistant professor, Princeton University

Visiting appointments and affiliations

2021– research professor, University of Copenhagen
2008–09 visiting researcher, Mathematical Institute, Utrecht
2005–06 fellow, Perimeter Institute for Theoretical Physics
2002– fellow, Pittsburgh Center for the Philosophy of Science

Education

2001 PhD philosophy, University of Pittsburgh
Locality, localization, and the particle concept: topics in the foundations of quantum theory
1999 visiting student researcher, Oxford University
1998 MA mathematics, University of Pittsburgh
1997 MA philosophy, University of Pittsburgh
1995 BA philosophy, with honors, Calvin College

Honors and prizes

· Behrman Fellow, Princeton Council of the Humanities, 2005–08
· James Cushing Memorial Prize in the History and Philosophy of Physics, 2004
· "Ten best articles in philosophy for the year 2002," *The Philosopher's Annual*
· "Best article by a recent PhD for the year 2001," Philosophy of Science Association
· "Ten best articles in philosophy for the year 2001," *The Philosopher's Annual*
· Alan Ross Anderson Fellowship for Philosophical Logic, University of Pittsburgh, 1997
· Pew Younger Scholars Graduate Fellow, 1995–98

Research funding

· Princeton Council of the Humanities grant, "The Modern Breakthrough in Scandinavia: Philosophy, Science, Art" (co-PI Bridget Alsdorf), 2024–26. $75,000
· John Templeton Foundation research grant, "Space from entanglement," 2024–26. $225,000

- Templeton World Charity Foundation research grant, "Experimental tests of quantum reality" (co-PI Andrew Briggs), 2013–16. $2,600,000
- National Science Foundation research grant, "Identical particles and statistics in superselection theory" (co-PI David J. Baker), 2011–12. $100,000
- Mellon New Directions in the Humanities Fellowship, 2008–09. $250,000
- John Templeton Foundation research grant, "Understanding the quantum world through mathematical innovation," 2007
- Princeton Tuck Grant for International Research Collaboration, 2018–19 (Denmark); 2010–11 (Netherlands)
- Princeton Center for the Study of Religion grant for developing the course "Religion and Scientific Objectivity," 2017
- Princeton 250th Anniversary Fund for Innovation in Undergraduate Education, grant for developing the logic curriculum, 2004

Publications

Books

3. *How logic works: a user's guide* (Princeton University Press, 2020), 10.1515/9780691208718.
2. *The logic in philosophy of science* (Cambridge University Press, 2019), 10.1017/9781316275603.
1. (with A. Briggs and A. Steane), *It keeps me seeking: the invitation from science, philosophy, and religion* (Oxford University Press, 2018), 10.1093/oso/9780198808282.001.0001.

Edited volumes

3. (with M. Ozawa, J. Butterfield, M. Rédei, Y. Kitajima, and F. Buscemi), *Reality and measurement in algebraic quantum theory* (Springer, 2018), 10.1007/978-981-13-2487-1.
2. *Deep beauty: understanding the quantum world through mathematical innovation* (Cambridge University Press, 2011), 10.1017/CBO9780511976971.
1. (with J. Butterfield), *Quantum entanglements: selected papers of Rob Clifton* (Clarendon Press, 2004), 10.1093/oso/9780199270156.001.0001.

Articles

37. (with J. Manchak and J. O. Weatherall), "Deterministic theories", Philosophy and Phenomenological Research (2026) 10.1111/phpr.70122.
36. "Fundamental physics and middle sized dry goods", Scientia et Fides (2025) 10.12775/SetF.2025.015.
35. (with J. Manchak, T. W. Barrett, and J. O. Weatherall), "Determinism and asymmetry in general relativity", British Journal for the Philosophy of Science (2025).
34. (with J. Manchak), "Closing the hole argument", British Journal for the Philosophy of Science **76**, 295–318 (2025) 10.1086/719193.
33. (with J. Butterfield), "John Bell on 'subject and object': an exchange", Journal for General Philosophy of Science **54**, 305–324 (2023) 10.1007/s10838-021-09594-y.
32. "What philosophy of science has to offer to theology", Philosophy, Theology, and the Sciences

10, 96–104 (2023) 10.1628/ptsc-2023-0009.

31. (with T. Barrett), "Mutual translatability, equivalence, and the structure of theories", Synthese **200**, 240 (2022) 10.1007/s11229-022-03733-8.
30. (with H. Kragh), "Cosmology and theology", Stanford Encyclopedia of Philosophy (2021).
29. "Concluding unscientific image", Metascience **29**, 175–185 (2020) 10.1007/s11016-020-00506-2.
28. (with D. Tsementzis), "Foundations and philosophy", Philosopher's Imprint **18** (2018).
27. (with T. Barrett), "From geometry to conceptual relativity", Erkenntnis **82**, 1043–1063 (2017) 10.1007/s10670-016-9858-y.
26. (with T. Barrett), "Quine's conjecture on many-sorted logic", Synthese **194**, 3563–3582 (2017) 10.1007/s11229-016-1107-z.
25. (with T. Barrett), "Glymour and Quine on theoretical equivalence", Journal of Philosophical Logic **45**, 467–483 (2016) 10.1007/s10992-015-9382-6.
24. (with T. Barrett), "Morita equivalence", Review of Symbolic Logic **9**, 556–582 (2016) 10.1017/S1755020316000186.
23. (with D. Baker and N. Swanson), "The conventionality of parastatistics", British Journal for the Philosophy of Science **66**, 929–976 (2015) 10.1093/bjps/axu018.
22. (with D. Baker), "How is spontaneous symmetry breaking possible? understanding Wigner's theorem in light of unitary inequivalence", Studies in History and Philosophy of Modern Physics **44**, 464–469 (2013) 10.1016/j.shpsb.2013.09.005.
21. "Plantinga on providence and physics", European Journal for Philosophy of Religion **5**, 19–30 (2013) 10.24204/ejpr.v5i3.216.
20. "Ruetsche on the pristine and adulterated in quantum field theory", Metascience **22**, 69–75 (2013) 10.1007/s11016-012-9702-1.
19. "The semantic view, if plausible, is syntactic", Philosophy of Science **80**, 475–478 (2013) 10.1086/671077.
18. "What scientific theories could not be", Philosophy of Science **79**, 183–206 (2012) 10.1086/664745.
17. (with D. Baker), "Antimatter", British Journal for the Philosophy of Science **61**, 93–121 (2010) 10.1093/bjps/axp009.
16. (with J. Bub), "Can quantum cryptography imply quantum mechanics?", Quantum Information and Computation **5**, 170–175 (2005) 10.26421/QIC5.2-8.
15. "Complementarity of representations in quantum mechanics", Studies in History and Philosophy of Modern Physics **35**, 45–56 (2004) 10.1016/j.shpsb.2003.01.001.
14. "On information-theoretic characterizations of physical theories", Studies in History and Philosophy of Modern Physics **35**, 277–293 (2004) 10.1016/j.shpsb.2004.02.001.
13. "Remote preparation of arbitrary ensembles and quantum bit commitment", Journal of Mathematical Physics **45**, 4920–4931 (2004) 10.1063/1.1812827.
12. (with R. Clifton), "No place for particles in relativistic quantum theories?", Philosophy of

Science **69**, 1–28 (2002) 10.1086/338939.

11. (with R. Clifton and J. Bub), "Characterizing quantum theory in terms of information-theoretic constraints", Foundations of Physics **33**, 1561–1591 (2002) 10.1023/a:1026056716397.

10. (with R. Clifton), "Are Rindler quanta real? inequivalent particle concepts in quantum field theory", British Journal for the Philosophy of Science **52**, 417–470 (2001) 10.1093/bjps/52.3.417.

9. (with R. Clifton), "Entanglement and open systems in algebraic quantum field theory", Studies in History and Philosophy of Modern Physics **32**, 1–31 (2001) 10.1016/s1355-2198(00)00033-2.

8. "On the nature of continuous physical quantities in classical and quantum mechanics", Journal of Philosophical Logic **30**, 27–50 (2001) 10.1023/a:1017574203443.

7. "Reeh-Schlieder defeats Newton-Wigner: on alternative localization schemes in relativistic quantum field theory", Philosophy of Science **68**, 111–133 (2001) 10.1086/392869.

6. (with R. Clifton), "Generic Bell correlation between arbitrary local algebras in quantum field theory", Journal of Mathematical Physics **41**, 1711–1717 (2000) 10.1063/1.533253.

5. (with R. Clifton and A. Kent), "Nonlocal correlations are generic in infinite-dimensional bipartite systems", Physical Review A **61** (2000) 10.1103/PhysRevA.61.042101.

4. "The Einstein-Podolsky-Rosen state maximally violates Bell's inequalities", Letters in Mathematical Physics **53**, 321–329 (2000) 10.1023/A:1007609031556.

3. (with R. Clifton), "Bipartite-mixed-states of infinite-dimensional systems are generically nonseparable", Physical Review A **61**, 012108 (1999) 10.1103/PhysRevA.61.012108.

2. (with R. Clifton), "Maximal beable subalgebras of quantum mechanical observables", International Journal of Theoretical Physics **38**, 2441–2484 (1999) 10.1023/A:1026628407645.

1. (with R. Clifton, D. Feldman, M. Redhead, and A. Wilce), "Superentangled states", Physical Review A **58**, 135–145 (1998) 10.1103/PhysRevA.58.135.

Chapters in books

20. "Carnap's formal philosophy of science", in *Carnap-Handbuch*, edited by C. Damböck and G. Schiemer (J.B. Metzler, Berlin, 2026), pp. 521–525, 10.1007/978-3-662-72937-3_82.

19. (with S. De Haro), "Scientific theories", in *Comprehensive philosophy of science*, edited by S. O. Hansson, Forthcoming (Elsevier, 2026).

18. "Bohr's epistemological lesson of quantum physics", in *How to understand quantum mechanics: 100 years of ongoing interpretations*, edited by J. Faye and L.-G. Johansson (Springer, 2026).

17. "The point of departure: Bohr's 'the unity of human knowledge (1960)'", in *Can we understand the world?*, edited by J. Hafner (Ergon Verlag, 2026).

16. "Niels Bohr: physicist-philosopher in action", in *La philosophie des physiciens, XXe-XXIe siècles*, edited by F. Laguens (2026).

15. "Rasmus Nielsen", in *Grundriss der geschichte der philosophie*, edited by G. Hartung (Schwabe Verlag, Basel, 2025).

14. (with A. S. Jacobsen), "Niels Bohr og den danske filosofiske tradition", in *Vitenskap, menneskerettigheter og diplomati: Fridtjof Nansen og Niels Bohr*, edited by J. Lothe and

H. Porsdam (Novus Forlag, Oslo, 2023), pp. 245–264.

13. "The philosophy of science in Either-Or", in *Kierkegaard's Either-Or*, edited by R. Kemp and W. Wietzke (Cambridge University Press, 2023), pp. 153–170, 10.1017/9781009067713.010.

12. "Objective description in physics", in *Proceedings of the 16th international congress of logic, methodology and philosophy of science and technology*, edited by T. Marvan, H. Andersen, B. Löwe, and H. Chang (College Publications, 2022), pp. 111–132.

11. "To be a realist about quantum theory", in *Quantum worlds: perspectives on the ontology of quantum mechanics*, edited by O. Lombardi (Cambridge University Press, 2019), pp. 133–163, 10.1017/9781108562218.010.

10. "A theological critique of the fine-tuning argument", in *Knowledge, belief, and God: new insights in religious epistemology*, edited by M. A. Benton, J. Hawthorne, and D. Rabinowitz (Oxford University Press, 2018), pp. 122–135, 10.1093/oso/9780198798705.003.0007.

9. (with D. Tsementzis), "Categories of scientific theories", in *Categories for the working philosopher*, edited by E. Landry (Oxford University Press, 2018), pp. 402–429, 10.1093/oso/9780198748991.003.0017.

8. "Why methodological naturalism?", in *The Blackwell companion to naturalism*, edited by K. J. Clark (Blackwell, 2016), pp. 136–149, 10.1002/9781118657775.ch10.

7. "Scientific theories", in *The Oxford handbook of philosophy of science*, edited by P. Humphreys (Oxford University Press, 2016), pp. 585–608, 10.1093/oxfordhb/9780199368815.013.33.

6. (with H. Kragh), "Physical cosmology", in *The Routledge companion to Theism* (Routledge, 2012), pp. 241–255, 10.4324/9780203123294.

5. "The measure of all things: quantum mechanics and the soul", in *The soul hypothesis: investigations into the existence of the soul*, edited by M. C. Baker and S. Goetz (Continuum Press, 2011), pp. 138–163.

4. (with M. Müger), "Algebraic quantum field theory", in *Handbook of the philosophy of physics*, edited by J. Butterfield and J. Earman (Kluwer Academic Publishers, 2006), pp. 731–864, 10.1016/b978-044451560-5/50011-7.

3. "Locality", in *The philosophy of science: an encyclopedia*, edited by J. Pfeifer and S. Sarkar (Routledge, 2006).

2. "Quantum mechanics", in *The philosophy of science: an encyclopedia*, edited by J. Pfeifer and S. Sarkar (Routledge, 2006).

1. (with R. Clifton), "Reconsidering Bohr's reply to EPR", in *Non-locality and modality*, edited by T. Placek and J. Butterfield (Kluwer, 2002), pp. 3–18, 10.1007/978-94-010-0385-8_1.

Book reviews

5. "On theories: logical empiricism and the methodology of modern physics", Mind **133**, 1161–1172 (2024) 10.1093/mind/fzac065.

4. "There are no such things as theories", Journal for General Philosophy of Science **52**, 609–612 (2021) 10.1007/s10838-021-09573-3.

3. "Platonism, naturalism, and mathematical knowledge", Notre Dame Philosophical Reviews (2013).

2. "On quanta, mind, and matter", Studies in History and Philosophy of Modern Physics **33**, 744–747 (2002) 10.1016/S1355-2198(02)00042-4.

1. "P. Vermaas, a philosopher's understanding of quantum mechanics: possibilities and impossibilities of a modal interpretation", British Journal for the Philosophy of Science **52**, 387–391 (2001) 10.1093/bjps/52.2.387.

Unpublished papers and notes

· "Invariance and ontology in relativistic physics" philpapers:HALTIN-3
· "Momentum and context" philpapers:HALMAC-2
· "Quantifier variance without collapse" philpapers:HALQVW
· "Against eliminating sorts" philpapers:HALAES-4
· "Mind independence and invariance"
· "Does quantum theory kill time?" philpapers:HALDQT
· (with S. Wolters) "Independence conditions for nets of local algebras as sheaf conditions" arxiv:1309.5639
· (with N. Swanson) "Comment on the structure of physics" philsci-archive:9314

Writing for wider audiences

· "Niels Bohr forvansket" Weekendavisen, Jun 2020
· "Can science speak about the past" Biologos, Dec 2017
· "Fine-tuning does not imply a fine tuner" Nautilus, Jan 2017
· "Matter" Edge Question 2017
· "Einstein was wrong" Edge Question 2016
· "What does quantum mechanics suggest about our perceptions of reality?" Big Questions Online 2015
· "When is belief in miracles rational?" Slate 2015
· "Meta-thinking" Edge Question 2015
· (with Adam Neder) "Quantum envy" Lutheran Forum 2007
· "Comments on Clouser's claims for theistic science" PSCF 2006

Invited lectures and conference presentations

2027 · "TBA" Physical and Philosophical Perspectives on von Neumann Algebras in Quantum Gravity, Amsterdam
· "Comments on David Wallace and Ted Sider" Eastern APA, symposium on Math-First Approaches to Physics and Metaphysics, Boston

2026 · "TBA" PSA 2026 symposium, New Perspectives on the Foundations of Quantum Field Theory, San Diego
· "TBA" Naturalized Metaphysics of Physics, Pittsburgh
· "Er musste das Wissen aufheben: van Fraassen on the priority of the manifest image" Schock Prize Symposium, Stockholm
· "Equivalence: A Program" Conference on theoretical equivalence, Oslo
· "Updating your religion" Reunions, Princeton

2025
- "Niels Bohr on causal explanation" 5th International Conference on the History of Quantum Physics
- "Theory acceptance is not (yet) belief" The epistemological significance of Kierkegaard's Climacus works, University of Southampton
- "Brandes against dualism" University of Copenhagen
- "Niels Bohr's philosophy of applied mathematics" Videnskabshistorisk Selskab, Copenhagen
- "Defining determinism" Workshop: The Logic in Science, Venice
- "When is a theory deterministic?" Radboud Center for Natural Philosophy, Nijmegen
- "Equivalence – state of play" Workshop: Università della Svizzera italiana
- "Reduction redux" Department Colloquium: Università della Svizzera italiana

2024
- "On the unity of science" Forum Religionen im Kontext, Potsdam
- "Mathematics is just more language" Meaning and Equivalence of Formal Theories, Vienna
- World Congress of Philosophy, Rome
- "Equivalence in foundations" ASL Logic Colloquium, Göteborg
- "Niels Bohr on the knowing subject" BOHR21 Workshop, Copenhagen
- "The roots of Niels Bohr's thought" Workshop on Literature, Philosophy, and Science (Transformative Transmissions: German-Scandinavian Intellectual Communities 1790–1860), Copenhagen

2023
- "Was Rasmus Nielsen a windbag (as Kierkegaard claimed)?" Søren Kierkegaard Research Center, Copenhagen
- "What do models say?" UC Irvine
- "Contextualism in relativity and quantum theories" University of Stockholm
- "Considerations from physics and theology" Fine Tuning Workshop, Rutgers
- "Speaking about models" University of Konstanz
- "Niels Bohr's values" University of Copenhagen
- "Dispensing with the hole argument" Mathematics in Physics, Oslo
- "Invariance and mind-independence" Theoretical Philosophy Seminar, University of Uppsala
- "The goal of formal analysis" The Formal Turn in Twentieth Century Thought, Vienna
- "The concept of a frame of reference" Physics meets Philosophy, Vienna
- "Spor efter Kierkegaard i den moderne fysik" Søren Kierkegaard Selskabet

2022
- "Niels Bohrs og den danske filosofiske tradition" Symposium: Niels Bohr: Scientist, Philosopher, and Diplomat, Copenhagen
- "Translations between translations" Tarskian Algebraic Logic, Relativity Theory and Methodology of Science – István Németi is 80, Alfréd Rényi Institute of Mathematics
- "Critical or dogmatic interpretation of quantum mechanics?" Grete Hermann and Friends, Utrecht
- "Mind independence" Cambridge-Copenhagen Symposium on Realism for Realistic People by Hasok Chang
- Three lectures on The Logic in Philosophy of Science: Winter school, University of Geneva

2020
- Central APA, Chicago
- Three lectures: Matematesering af Fysik, for Videnskabsteori course, University of Copenhagen

2019
- "Physics and the absolute conception" University of Illinois
- "Objective description in physics" The International Congress of Logic, Methodology and Philosophy of Science and Technology, Prague

· Pacific APA, Vancouver

2018 · "2-categories of theories" Conference on Category Theory in Philosophy of Science, LMU Munich

· "Semantics and the god's eye view" Conference on Semantics of Science, University of Salzburg

· "Wissenschaftslogik erneuert" LMU Munich

2017 · "The wave-function in itself" Conference: Identity, indistinguishability and non-locality in quantum physics, Buenos Aires

· "The invariant content of equivalent theories" CSLI Workshop on Logic, Rationality, and Intelligent Interaction, Stanford University

· "Probability ex nihilo" Rutgers Probability Seminar

2016 · "The invariant content of equivalent theories" Semantics of Scientific Theories Conference, LMU Munich

· "Probability and fine-tuning" Multiverse and Theodicy Conference, Rutgers

· "From geometry to conceptual relativity" (with T. Barrett) Society for Exact Philosophy Conference, Miami

2015 · "The categorical approach to scientific theories" Notre Dame

· "Glymour and Quine on theoretical equivalence" (with T. Barrett) Logic, Relativity and Beyond Conference, Budapest

· "Divine and human agency in a quantum world" Science and Personal Action Conference, MIT

· "What's wrong with the fine-tuning argument?" New Insights and Directions in Religious Epistemology Conference, Oxford University

· "The invariant structure of equivalent theories" Rutgers University Philosophy Colloquium

· "Reality and equivalence in algebraic quantum theory" Nagoya Winter Workshop 2015: Reality and Measurement in Algebraic Quantum Theory, University of Nagoya, Japan

2014 · "Univalent foundations: structuralist foundations?" (with D. Tsementzis) Association of Symbolic Logic Meeting, Philadelphia

· "Categories of scientific theories" Philosophy of Science Association Meeting, Chicago

· "Equivalent theories and invariant content" University of Southern California Philosophy Colloquium

· "Explanation via surplus structure" Sigma Club, London School of Economics

· "Does the universe need God?" Ian Ramsey Centre for Science and Religion, Oxford University

· "Does the universe need God?" Faraday Institute Lecture Series, Cambridge University

2013 · "Quantum theory of infinite systems: from fields to information" Instituto de Física Fundamental, Madrid

· "What is a scientific theory?" (with J. Weatherall) Carnegie Mellon University

· "The conventionality of parastatistics" (with D. Baker and N. Swanson) Foundations of Physics, Munich

· "The invariant structure of equivalent theories" keynote lecture at LMP graduate conference, University of Western Ontario

· "Structuralist foundations for abstract mathematics" (with D. Tsementzis) Category-theoretic foundations of mathematics workshop, UC Irvine

· "The relativity of wrong, redux" Philosophy, Science and Religion Forum, CUNY, Laguardia Community College

· "What does physics have to do with theology anyway?" The Walton Lecture in Science, Philosophy,

and Religion, Fordham University

· "Locality conditions as universal properties" (with S. Wolters) Conference on Relativistic Causality, University of Pittsburgh

2012 · "What is a scientific theory?" Philosophy of Science Association Meeting, San Diego

· "Ruetsche on the pristine and adulterated in quantum field theory" Rotman Institute for Philosophy, University of Western Ontario

· "Ontological naturalism and the interpretation of quantum mechanics" PhysPhil Conference, St. Andrews University, Scotland

· "Philosophy of mind meets quantum mechanics" Summer School in Philosophy of Religion, University of St, Thomas, Minneapolis

· "Divine action in a quantum world?" Faith, philosophy of science, and science, Calvin College

2011 · "What scientific theories could not be" University of Washington, Seattle

· "Does physical cosmology confirm theistic belief?" McIntyre Lecture on the Foundations of Physics, Wheaton College, Illinois

· "Foundations for categories, by categories" Foundational Questions in the Mathematical Sciences, Traunkirchen, Austria

· "Natural structure on state space" Philosophy of Physics Workshop, UC Irvine

· "What scientific theories could not be" LPS Colloquium, UC Irvine

2010 · "Ontological naturalism and the interpretation of quantum mechanics" Philosophy Colloquium, Beijing University

2009 · "Recent results on duality for the category of first-order theories" Mathematical Logic Seminar, University of Nijmegen

· "Notions of monoidal category" Operads Seminar, University of Utrecht

2007 · "Antimatter and the metaphysics of opposites" University of Texas at Austin

2006 · "Representing non-local boxes with C^*-algebras" Perimeter Institute for Theoretical Physics

· "On the conventionality of the claim, 'There are para-particles' " Minnesota Center for the Philosophy of Science

· "Otherworldly information theory" Boston Colloquium for Philosophy of Science

· "Deriving quantum mechanics from information-theoretic axioms" Conference in honor of John Wheeler, Princeton

2005 · "No reductive physicalism, no measurement problem" New Directions in the Foundations of Physics, College Park, Maryland

· "No reductive physicalism, no measurement problem" University of Southern California

2004 · "Algebraic quantum field theory" Center for Philosophy of Science, University of Pittsburgh

· "No reductive physicalism, no measurement problem" University of Notre Dame

· "A generalized no-bit-commitment theorem for quantum systems" C^*-algebras and quantum information theory, UC Santa Barbara

· "Information-theoretic axioms for quantum mechanics" Contemporary issues in philosophy of physics, University of Western Ontario

2003 · "Can quantum cryptography imply quantum mechanics?" University of Maryland

· "Can the quantum eraser undo the past?" Faculty Seminar on the Concept of Time, Princeton Council of the Humanities

· "Characterizing quantum theory in terms of information-theoretic constraints" Philosophy of

Physics Workship, Oxford University

2002 · "Bohr versus Bohm on ontological pluralism" Philosophy of Science Association Meeting, Milwaukee

· "Complementarity in quantum field theory" Boston Colloquium for Philosophy of Science

· "Objective indeterminacy: A prolegomenon to Niels Bohr's philosophy of quantum theory" Princeton Philosophical Society

· "Non-existent quantities in quantum mechanics" New directions in the foundations of physics, College Park, Maryland

· "Complementarity of representations in quantum mechanics" Philosophy of mathematics in application, UC Irvine

2001 · "Reconsidering Bohr's reply to Einstein-Podolsky-Rosen" Modality, probability, and Bell's theorem, Jagiellonian University, Poland

· "Complementarity of representations in quantum mechanics" International Quantum Structures Association Meeting, Cesena, Italy

· "The 'substance' of complementarity: Niels Bohr, quantum field theory, and particle metaphysics" Princeton University; University of Notre Dame; University of Michigan

2000 · "Does relativity imply nonlocality? Unpacking the Reeh-Schlieder theorem" UC Irvine

1999 · "Generic Nonlocality in Theories of Everything" (with R. Clifton) London School of Economics

· "Bell correlations in the vacuum: on a conjecture of David Malament" (with R Clifton) Oxford University

Public lectures and outreach

· "Niels Bohr's great idea" BLOOM festival of science and ideas, Copenhagen. May 2022

· "Entydig Videnskab: Niels Bohrs Aktualitet i Fysik og Filosofi" University of Copenhagen 542nd Anniversary Lecture. Nov 2021

· "Reason, science, and mystery," and "Learning theology in a new landscape." Festival of Learning, Svenska Kyrkan. 2017

· Panelist at "The physics of everything," New York Academy of Sciences, Apr 2016

· Panelist at "Liberal arts education, religious life, and the Jewish tradition" Jewish Learning Initiative on Campus Princeton, 2014

· Veritas forums at Brown, CalTech, NYU, Washington U, Rider, Columbia. 2014–17

· The Ratio Project paper development workshop for philosophers in Belarus, Russia, and the Ukraine. Moscow, July 2014

· Academic–Athletic Fellow, Princeton University Rowing. 2011–

· Philosophy of religion summer seminar, University of Beijing, 2010

Teaching and advising

Doctoral and postdoctoral advising

I have worked with and written letters for the PhD students and postdocs listed below — ten of them as primary dissertation advisor, marked with an asterisk. Where a student's PhD is from another university, that university is given in parentheses. Unless otherwise indicated, the placement listed is a tenure-track position.

· Jake Khawaja*

· Brendan Kolb*
· Alexander Meehan*: postdoc Yale, University of Wisconsin at Madison
· David Schroeren*: postdoc Geneva
· Daniel Berntson: postdoc Rutgers, postdoc Uppsala
· Robbie Hirsch*: postdoc Princeton
· Thomas Barrett*: postdoc NYU, UC Santa Barbara
· Dimitris Tsementzis*: postdoc Rutgers, Goldman Sachs machine learning quants
· Laurenz Hudetz (Salzburg): London School of Economics
· Neil Dewar (Oxford): postdoc Munich, Cambridge University
· Michaela McSweeney: Boston University
· Benjamin Feintzeig (UC Irvine): University of Washington
· Noel Swanson*: University of Delaware
· Samuel Fletcher (UC Irvine): University of Minnesota
· Nicholas Teh (Cambridge): University of Notre Dame
· Joseph Rachiele*: Chapman University
· Joshua Hershey: The King's College NY
· Jo E. Wolff (Stanford): University of Edinburgh
· Jim Weatherall (UC Irvine): UC Irvine
· Caleb Cohoe: Metropolitan State University, Denver
· Jada Strabbing: Fordham University, Wayne State University
· Giovanni Valente (Maryland): University of Pittsburgh, Politecnico Milano
· Yuichiro Kitajima (Hokkaido): Nihon University
· David J. Baker*: University of Michigan
· Lara Buchak: UC Berkeley, Princeton
· Tracy Lupher (Texas): James Madison University
· Antony Eagle: Oxford University, University of Adelaide

Students in Princeton's PhD program earn "units of credit" by writing papers under the supervision of a member of the faculty. I have advised unit work in general philosophy of science, philosophy of physics, philosophy of religion, and philosophical logic.

Graduate seminars

· Spacetime and objectivity
· Kierkegaard's *Concluding Unscientific Postscript*
· Philosophy of space and time (with D. Hogan)
· Logical philosophy of science
· Foundations of mathematics: set theory vs. category theory (with J. Burgess)
· Categorical logic and topos theory
· Quantum information theory and the foundations of quantum mechanics (with B. van Fraassen)
· Metaphysics of physics (with J. Butterfield)
· Foundations of quantum field theory
· Scientific realism and antirealism
· From physics to metaphysics (with A. Elga)

Undergraduate courses and seminars

- Religion and Scientific Objectivity
- Kierkegaard
- Philosophy of Science
- History of Analytic Philosophy 1900–1950
- Philosophy of Mathematics
- Category Theory (math–phil)
- Advanced Logic (math–phil)
- Intermediate Logic
- Philosophical Logic
- Philosophy of Physics
- Philosophy of Religion
- Introductory Logic
- History of Modern Philosophy
- The Modern Breakthrough in Scandinavia

Senior theses

Since 2001, between one and five theses per year, for Philosophy, Mathematics, Physics, and ORFE. Besides my areas of expertise, I have advised theses in pragmatism, metaphysics, philosophy of mind, religion, biology, and medicine.

Academic service

University and departmental administration

- Princeton Committee on Undergraduate Admission and Financial Aid, 2024–27
- Princeton Policy Committee on Athletics and Campus Recreation, 2022–25
- Princeton University Research Board, 2007–08, 2009–11, 2016–17
- Department of Philosophy:
  - Director of Graduate Studies, 2007–08
  - appointments committee, 2018–20
  - director of PhD placement, 2009–12, 2016–17, 2022–
  - director of colloquium committee, 2010–11, 2016–17
  - graduate admissions committee, 2001–02, 2003–06, 2007–08, 2009–11, 2013–14
  - curriculum committee, 2001–03, 2004–06, 2007–08
  - computer committee, 2004–06, 2014–15
  - library committee, 2009–10
  - graduate language examiner (German), 2001–

Editorial work

- Associate editor, *Philosophy of Physics*, 2022–
- Associate editor, *Review of Symbolic Logic*, 2020–
- Associate editor, *Science, Religion and Culture*, 2017–
- Area editor for philosophy of physical science, PhilPapers.org, 2011–

Conference organization and program committees

- Co-organizer, Oxford–Princeton partnership in philosophy of physics, 2001–07
- Organizer, Conference on understanding quantum mechanics through mathematical innovation, Princeton, Oct 2007
- Co-organizer, Symposium on categorical perspectives in the philosophy of physics, Philosophy of Science Association Meeting, 2010
- Co-organizer, Irvine–Pittsburgh–Princeton conference series on the foundations of physics, 2013–15
- Organizer, Conference on equivalent theories in science and metaphysics, Princeton, Mar 2015
- Co-organizer, Foundations of Quantum Theory Conference, Nagoya, Japan, Mar 2015
- Co-organizer, Workshop on Georg Brandes and the Modern Breakthrough, May 2025
- Co-organizer, Workshop on Kierkegaard, Objectivity, and Subjectivity, Jul 2025
- Program committee:
  - Philosophy of Science Association Meeting, 2006
  - The semantics of scientific theories, LMU Munich, 2016
  - European Philosophy of Science Association Meeting, 2023
  - Foundations of Physics, 2026
  - Foundations of Physics, Nanyang Technological University, Singapore, 2027

External evaluation and advisory work

- Scientific Advisory Board, ERC Advanced Grant project "Team semantics and dependence" (PI: Jouko Väänänen), 2021–24
- External reviewer for faculty hiring and promotion: Boston University, Chicago, Columbia, CUNY, CCNY, Drexel, Georgetown, London School of Economics, National Technical University of Athens, Notre Dame, NYU, Oxford, Radboud University of Nijmegen, Rutgers, St. Olaf College, Stanford, UC Irvine, University of Chicago, University of Edinburgh, University of Michigan, Pittsburgh, University of Southern California, University of Vienna, University of Washington, University of Waterloo, Wheaton College, Yale
- External dissertation examiner: University of Geneva; University of Texas; University of Utrecht (foundations of science); Radboud University of Nijmegen (mathematics); Carnegie Mellon University; LMU Munich; University of Aberdeen (theology); Roskilde Universitet; University of Salzburg
- Academic Advisory Board, National Speech and Debate Association, 2015–16
- Invited participant, New York Institute of Philosophy, cosmology and religion discussion group, 2008
- Invited participant, Princeton Council of the Humanities, philosophy of time group, 2003–04

Refereeing

- Journals and conferences: *Analysis*, *Annals of Physics*, *Annals of Pure and Applied Logic*, *Australasian Journal of Philosophy*, *British Journal for the Philosophy of Science*, *Bulletin of Symbolic Logic*, *Canadian Journal of Philosophy*, *Christian Scholars Review*, *Communications in Mathematical Physics*, *Erkenntnis*, *European Journal for Philosophy of Science*, *Foundations of Physics*, *International Journal for Philosophy of Religion*, *International Studies in the Philosophy of Science*, *Journal for General Philosophy of Science*, *Journal of Logic & Analysis*, *Journal of Mathematical Physics*, *Journal of Philosophy*, *Journal of Physics A*, *Logic and Logical*

Philosophy, *Notre Dame Journal of Formal Logic*, *Nous*, *Philosophia Mathematica*, *Philosopher's Imprint*, *Philosophical Studies*, *Philosophical Transactions of the Royal Society*, *Philosophy and Phenomenological Research*, *Philosophy Compass*, *Philosophy of Science*, *Physical Review A*, *Physical Review Letters*, *Physics Letters A*, *Quantum Studies: Mathematics and Foundations*, *Ratio*, *Religious Studies*, *Reviews of Mathematical Physics*, *Science and Christian Belief*, *Studia Logica*, *Studies in History and Philosophy of Modern Physics*, *Studies in History and Philosophy of Science*, *Theoria*; Quantum Physics and Logic (QPL 2011).

- Funding agencies: National Science Foundation (USA); Engineering and Physical Sciences Research Council (UK); Israel Science Foundation; Netherlands Organisation for Scientific Research (NWO); Royal Netherlands Academy of Arts and Sciences (KNAW); Social Sciences and Humanities Research Council of Canada; Agence Nationale de la Recherche (France); Austrian Science Fund (FWF); German-Israeli Foundation for Scientific Research and Development; European Research Council; National Research, Development and Innovation Office of Hungary; Foundation for Polish Science; The Leverhulme Trust; John Templeton Foundation; Templeton World Charity Foundation.
- Publishers: Elsevier; Hackett; Oneworld; Oxford University Press; Princeton University Press; Springer; Wiley-Blackwell; Yale University Press.

Languages

- English, native speaker
- Danish, C1 (Studieprøven)
- German, C1 (Zentrale Mittelstufenprüfung)
- Dutch, B2
- French, reading

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